For grading purposes	
Problem 1	
Problem 2	
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Studio	
Total:	

Physics 214/224, Quiz 4

Primary instructor: Yurii Maravin

04/14/2022

Key

WID:__________________________________ NAME:____

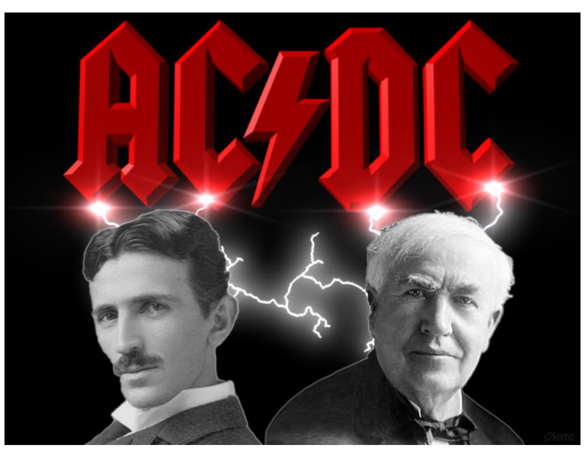
Instructions. Print and sign your name on this quiz and on your scantron card. In doing so, you are acknowledging the KSU Honor Code: "On my honor as a student I have neither given nor received unauthorized aid on this academic work."

For two quick points, circle your studio instructor:

Lohman (TU 7:30) Lohman (TU 9:30) Maravin (TU 11:30)

Kumarappan (TU 1:30) Blaga (FW 9:30) Maravin (FW 11:30)

Work alone and answer all questions. Part I questions must be answered on the Scantron cards. Put your name on your card. Color in the correct box completely for every answer with a pencil. If you make a mistake, erase thoroughly. **Don't forget to color in the boxes for your WID. If we have to correct this by hand we may take off five points from your score!** Part II must be answered in the space provided on the test. The last page is a detachable equation sheet. You may use a calculator. You may ask the proctors questions of clarification. Show your work in a clear and organized manner. You may receive partial credit for partial solutions. Solutions lacking supporting calculations will receive no points.



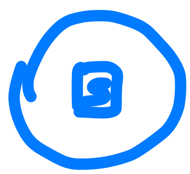
Interesting trivia: did you know that when listening to AC/DC, Nicola Tesla was **only** listening to AC?

Part I: All questions to be answered on your Scantron card (34 points total)

For the next three questions, indicate the direction of induced current flow for the following situations. Each involves a single circular loop of wire and a bar magnet. The plane of the wire loop is parallel to the floor and viewed from above, so that its current is either clockwise or counterclockwise. The bar magnet is always positioned with its north-south line perpendicular to the plane of the loop.

1. (2 points) The magnet is moved south-pole-first from a position below the coil towards the center of the coil. Coil is fixed.

   (a) Clockwise.
   (b) Counterclockwise.
   (c) No current will flow.



2. (2 points) The magnet is held fixed with the plane of the fixed coil half-way between north and south poles.

   (a) Clockwise.
   (b) Counterclockwise.
   (c) No current will flow.

3. (2 points) The coil is moved up away from the closer north pole of the magnet.

   (a) Clockwise.
   (b) Counterclockwise.
   (c) No current will flow.

4. (2 points) Which statement is most correct for capacitors operating in AC circuits?

   (a) Capacitors are short circuits at low frequencies and open circuits at high frequencies.
   (b) Capacitors are short circuits at high frequencies and open circuits at low frequencies. $\frac{1}{\omega C}$
   (c) Capacitors impede the current the same way at all frequencies.
   (d) Capacitors cannot operate in AC circuits.

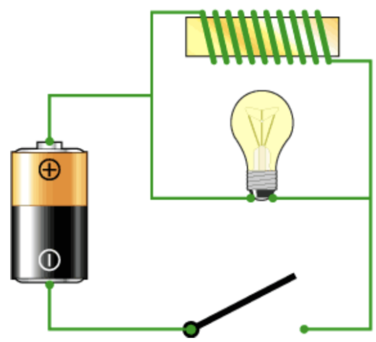
5. (2 points) Which statement is most correct for inductors operating in AC circuits?

   (a) Inductors are short circuits at low frequencies and open circuits at high frequencies. ωL
   (b) Inductors are short circuits at high frequencies and open circuits at low frequencies.
   (c) Inductors impede the current the same way at all frequencies.
   (d) Inductors cannot operate in AC circuits.

6. (2 points) Which statement is most correct for resistors operating in AC circuits?

   (a) Resistors are short circuits at low frequencies and open circuits at high frequencies.
   (b) Resistors are short circuits at high frequencies and open circuits at low frequencies.
   (c) Resistors impede the current the same way at all frequencies.
   (d) Resistors cannot operate in AC circuits.

7. (2 points) In the circuit below, the switch has remained open for a long time, and the LR time constant of the circuit is 30 s. Assume the inductor to be ideal, i.e., neglect its resistance. Just after the switch is closed:



   (a) The light will remain dim.
   (b) The light would come on and continue to stay on with a constant brightness.
   (c) The light would come on and then begin to dim gradually.
   (d) The light would briefly flash and then immediately turn off.
   (e) The circuit will explode!

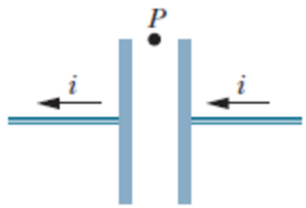
8. (2 points) Referring to the same diagram as the previous problem, if we waited for a very long time with the switch closed and then touched the solenoid and the light bulb

   (a) Both the solenoid and the lamp would feel very hot.
   (b) Both the solenoid and the lamp would feel cold.
   (c) The solenoid would feel hot while the lamp would feel cold.
   (d) The solenoid would feel cold while the lamp would feel hot.

9. (2 points) Referring to the same diagram again, if we waited a long time with the switch closed and then re-opened the switch

   (a) The light would remain lit.
   (b) The light would remain off.
   (c) The light would come on brightly and then begin to dim gradually.
   (d) The light would briefly flash and then immediately turn off.

The figure below shows a circular parallel-plate capacitor and the current in the connecting wire that is **charging** the capacitor. In the next three questions you need to identify correct statements regarding the electric field and its flux between the capacitor's plates, direction of the displacement current, and the direction of magnetic field.



10. (2 points) Select the correct statement below:

    (a) The direction of the displacement current i_d between the plates is leftward.

    (b) The direction of the displacement current i_d between the plates is rightward.

    (c) The displacement current is zero, i.e., no direction.

    (d) The displacement current changes its direction from leftward to rightward.

    (e) Impossible to determine.

11. (2 points) Select the correct statement below:

    RHR

    (a) The magnetic field at point P is directed into the page.

    (b) The magnetic field at point P is directed out of the page.

    (c) There is no magnetic field at point P because there is no actual current of charged particles between the plates.

12. (2 points) If the current, used to charge the capacitor is 2 A, and if the area of each plate is 1 m^2, the displacement current through a 0.2 m^2 area axially centered and wholly between the capacitor plates and parallel to them is

    (a) 2 A

    (b) 1 A

    $\frac{0.2}{1} \cdot 2\text{A} = 0.4\text{ A}$

    (c) 0.6 A

    (d) 0.4 A

    (e) 0.2 A

13. (2 points) When you decrease the frequency of the oscillator in a series RLC circuit by half:

    (a) the capacitive reactance is doubled

    $X_C = \frac{1}{\omega C}$ $X_L = \omega L$

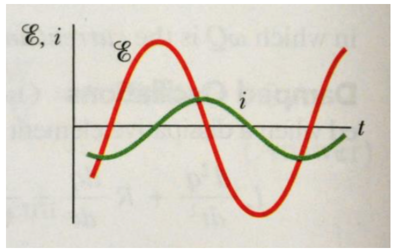
    (b) the capacitive reactance is halved

    (c) the impedance is doubled

    (d) the current amplitude is doubled

    (e) the current amplitude is halved

14. (2 points) In the figure below, the circuit's load is



$\mathcal{E} L i$

(a) mainly capacitive
(b) mainly inductive
(c) none of the above

15. (2 points) Light is called polarized if

(a) The direction of propagation of the waves is always at 90 degrees to the directions of the electric field vectors $\vec{E}$.
(b) The direction of the electric field vectors $\vec{E}$ of the waves and the direction of the wave propagation are always in the same plane.
(c) The direction of the magnetic field vectors $\vec{B}$ is at 90 degrees to the direction of the electric field vectors $\vec{E}$ of the waves.
(d) The direction of propagation of the waves is always at 90 degrees to both the directions of $\vec{E}$ and $\vec{B}$.
(e) it is about to go through a fancy Ray-Ban polarized sunglasses.

In the last two questions, unpolarized light with intensity I_0 passes sequentially through two polarizers with transmission axes oriented at 0 and 45 degrees, respectively.

16. (2 points) The intensity of the light leaving the first polarizer relative to the incident intensity is

(a) $I_1/I_0 = 0$.
(b) $I_1/I_0 = 1/4$.
(c) $I_1/I_0 = 1/\sqrt{2}$.
(d) $I_1/I_0 = 1/2$.
(e) $I_1/I_0 = 2$.

$$I_1 = \langle I_0 \cos^2\varphi \rangle = \frac{I_0}{2}$$

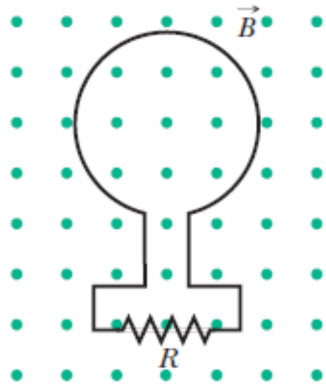
17. (2 points) The intensity of the light leaving the second polarizer relative to the incident intensity is

(a) $I_1/I_0 = 0$
(b) $I_1/I_0 = 1/4$.
(c) $I_1/I_0 = 1/\sqrt{2}$.
(d) $I_1/I_0 = 1/2$.
(e) $I_1/I_0 = 2$.

$$\frac{I_2}{I_1} = \cos^2 45 = \frac{1}{2}$$

$$\frac{I_0}{I_2} = \frac{I_0}{I_1} \cdot \frac{I_1}{I_2} = 4$$

Part II: Work out this part in the space provided. Show your work!
(39 points total)



18. (10 points) The magnetic flux through the loop shown in the figure above has the following time dependence $\Phi_B = 2.0t^2 - 3.2t$, where Φ_B is in milliwebers and t is in seconds. The magnetic field is pointing out of the page.

(a) (2 points) Write down a formula that describes an EMF in the circuit at any given time ($t > 0$ s) using the given information.

$$\underbrace{\mathcal{E} = -\frac{d\Phi_B}{dt}}_{+1} = -\frac{d}{dt}(2t^2 - 3.2t) = \underbrace{-4t + 3.2}_{+1}$$

NB. Absolute value is ok ($\mathcal{E} = |4t - 3.2|$)

(b) (2 points) What is the magnitude of the EMF induced in the loop when $t = 2$ s?

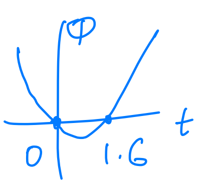
$-4 \cdot 2 + 3.2 = -4.8 \Rightarrow$ EMF $= 4.8$ V (+1)

Correct sign ("+") (+1)

NB Do not remove points if a formula in (a) is incorrect, but correctly applied here.

(c) (~~3~~ 4 points) What is the direction of the current through R (left, right, or no current?) at time $t = 2$ s? Explain your answer. (*Hint: is flux increasing or decreasing around $t = 2$ s?*)

$2t^2 - 3.2t = t(2t - 3.2)$



At $t = 2$s flux is increasing (+1)
(Also $d\Phi/dt > 0$ at $t = 2$) Explanation (+1)

Use RHR to determine the direction of the current to induce B into the page $\rightarrow$ CW $\Rightarrow$ left (+1)

Expl. (+1)

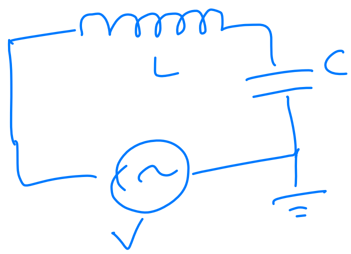
(d) (~~3~~ 2 points) Find a time when there is no current flowing through the resistor R.

$i = 0$ if $\mathcal{E} = 0$ (+1) $\Rightarrow -4t + 3.2 = 0 \Rightarrow$

$t = \frac{3.2}{4}$ s $= 0.8$ s (+1)

19. (11 points) A long piece of wire is connected to one side of an AC power source. One thousand turns of the wire are then wound around a 20 cm long, 3 cm radius cylindrical piece of iron with relative permeability $\mu_R = 600$. The same wire is then connected to a 20 cm^2 area flat conductor. A second identical piece of the conductor is positioned parallel to and 1 mm away from the first piece and is connected back to the other side of the power supply and grounded.

(a) (3 points) Draw schematics of the circuit.



L, C, V indicated? (+1)

in series? (+1)

ground indicated? (+1)

(b) (3 points) Calculate the inductance of the inductor.

$$L = \mu_0 \mu_R \frac{N^2}{\ell} A = 4\pi \cdot 10^{-7} \cdot 600 \cdot \frac{(1000)^2}{0.2} \cdot \pi (0.03)^2 \, H =$$

(+1) (+1)

$$= 10.7\, H$$ (+1)

(c) (2 points) Calculate the capacitance of the capacitor.

$$C = \varepsilon_0 \frac{A}{d} = 8.85 \cdot 10^{-12} \cdot \frac{20 \cdot (0.01)^2}{10^{-3}} F = 1.77 \cdot 10^{-11} F$$

(+1)

$$C = 17.7\, pF$$ (+1)

(d) (3 points) What operational frequency of the AC power supply will result in the maximal heating of the long piece of wire?

Heating is maximized when the current in the circuit is maximal. All thing being equal, it happens when Z is minimal:

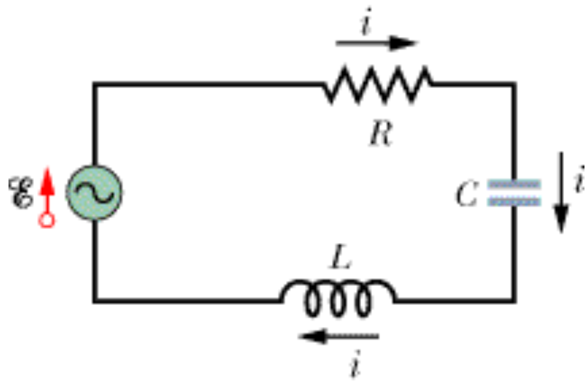
Explanation (+1)

$$i = \frac{\mathcal{E}}{Z} \Rightarrow \quad Z \text{ is minimal @ resonance}$$

(+1)

$$\omega = \frac{1}{\sqrt{LC}} = 72700 \frac{rad}{s} \quad \text{or} \quad f = \frac{\omega}{2\pi} = 11.5 \text{ kHz}$$

(+1)



20. (12 points) In the RLC circuit above assume that $R = 10.0\ \Omega$, $L = 20$ mH, $C = 0.6\ \mu$F, and the power source provides an RMS voltage $\mathcal{E} = 120$ V at a frequency f that can be varied.

(a) (4 points) For frequency $f = 60$ Hz, calculate impedance Z of the circuit.

+1 $Z = \sqrt{R^2 + (X_c - X_L)^2}$ $X_C = \frac{1}{\omega C} = \frac{1}{2\pi f C}$ +1 $= 4420\ \Omega$

$X_L = \omega L = 2\pi f L$ +1 $= 7.5\ \Omega$

$Z = 4410\ \Omega$

+1

(b) (3 points) At some frequency f_0, an amplitude of the current through the resistor is maximal. What is this frequency?

i is maximal when Z is minimal $\Rightarrow$ $\underline{X_C = X_L}$

+2 $\underline{\text{resonance}}$ $\omega = \frac{1}{\sqrt{LC}} = 9130\ \frac{rad}{s}$ or $f_0 = \frac{\omega}{2\pi} = 1450\ Hz$

or +1

(c) (3 points) What is the impedance Z of the circuit for that frequency, f_0, and how does it compare with R?

If $X_C = X_L \Rightarrow Z = R = 10\ \Omega$ Explanation +1

+2

(d) (2 points) If you double the frequency of the power source to $2f_0$, what happens to

i. capacitive reactance of the circuit? (increases, decreases, stays the same) +1

ii. inductive reactance of the circuit? (increases, decreases, stays the same) +1

(For this bullet point, no need to provide an explanation, just circle the appropriate answer)

$\frac{1}{\omega C}$ ωL

21. (6 points) TL;DR You have a light with a wavelength $\lambda_{in} = 128$ nm going into a wavelength shifter. The light that comes out of the shifter has a frequency 3.5 times smaller than the input and the intensity is half as what it used to be.

A long description of the problem, you don't have to read it:
Scintillator is a material that fluoresces when struck by a charged particle or high-energy photon. Scintillator detectors are very common, used in medical devices (PET scanners), high energy physics experiments (such as the one Dr. Maravin is working on), and in many nuclear and homeland security applications. One of such scintillators is a liquid argon (LAr). When an ionizing particle hits the LAr scintillator, the resulting scintillation light peaks at 128 nm, i.e., in deep ultraviolet spectrum. To detect this light, one needs to have a photosensitive detector that can efficiently detect such a low wavelength light. Unfortunately, none of the existing photodetector technologies are optimized for such low wavelength. To alleviate this problem, so-called wavelength shifters are used.

To convert the scintillating light to a light that the photosensors can detect, an organic compound, 1,1,4,4-tetraphenyl-1,3-butadiene (TPB) is commonly used. It absorbs LAr light and re-emits with a frequency that is 3.5 smaller than that of the original LAr scintillating light. The emitted light is also dimmer, with the intensity of the re-emitted light is about twice as small as the original LAr light.
The end of the long description.

(a) (3 points) What is the wavelength of the re-emitted light?

Expl. +1

$\lambda = \dfrac{c}{f} \Rightarrow$ if f gets smaller, $\lambda \to$ gets bigger

$\lambda_{out} = 128\,\text{nm} \cdot 3.5 = 448\,\text{nm}$ +2

(b) (3 points) What is a ratio of the magnetic field amplitude B of the re-emitted light to that of the LAr one?

Expl. +1

Intensity $\sim$ amplitude square (either $\vec{E}$ or $\vec{B}$)

$\dfrac{I_{out}}{I_{in}} = \dfrac{1}{2} = \dfrac{B_{out}^2}{B_{in}^2} \Rightarrow \dfrac{B_{out}}{B_{in}} = \dfrac{1}{\sqrt{2}} = 0.71$

$B_{out} = 0.71\,B_{in}$ +2